RV College of Engineering
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

2025 - Imp CIE

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USN

Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**17th January 2025**	**Maximum Marks**	**10 + 50**
Course Code	**21HS71**	**Duration**	**120 Mins**
Sem	**VII Semester**	**Improvement CIE**	
CONSTITUTION OF INDIA AND PROFESSIONAL ETHICS			

Note:
1. Answer all the Questions.

PART – A

Sl. No.	Questions	M	BT	CO
1	_____is termination by the employer of the service of a worker for any reason whatsoever, otherwise than as a punishment inflicted by way of disciplinary action.	01	2	2
2	The _____Act provides certain benefits to employees in case of sickness, maternity and employment injury, related matters.	01	1	2
3	The working hours of an adolescent worker shall be regulated in accordance with the provisions of the _____Act.	01	2	2
4	List any two Roles Played By the CSR Committee	01	2	2
5	List any two objectives of Industrial policy in India	02	2	2
6	What is child labor?	02	1	3
7	What is Fault Tree Analysis Approach?	02	3	1

PART – B

Sl. No.	Questions	M	BT	CO
1.	Define Corporate Social Responsibility? Explain the role of and responsibilities of CSR in society with two cases.	10	2	4
2.	Discuss the Guidelines stipulated for Universities and colleges to curb the menace of ragging.	10	1	3
3	Explain the prohibition and prevention of sexual harassment of women employee in working place with decided case study.	10	3	3
4	Highlight the thrust areas of Government in framing labour laws and list various labour legislation laws enacted by Central Government.	10	2	2
5	Define Industrial dispute. Explain the mechanism for resolving Industrial disputes under the Industrial dispute Act 1947.	10	2	4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max Marks	Quiz	--		--	--	--	--	10	--	--	--	--
		Test	--	10	20	20	--	10	30	10	-	-	--

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**17th February 2025**	**Maximum Marks**	**10 + 50**
Course Code	**21IM72**	**Duration**	**120 Mins**
Sem	**VII Semester**	**Improvement CIE**	

PRODUCT DESIGN & DEVELOPMENT

Note:

1. Answer all the Questions.
2. Avoid verbose answers and answer specifically and to the point.
3. Sketches must be provided in Part B answers as appropriate.
4. Engineering specifications and Qualitative and/or Quantitative metrics must be mentioned in Part B answers appropriately.

PART – A

Sl. No.	Questions	M	BT	CO
1	What is the role of Monte Carlo simulation in risk analysis for product development projects?	01	L_2	CO4
2	How does Design Structure Matrix (DSM) help in managing dependencies in iterative product design projects?	01	L_2	CO4
3	Name one key risk associated with product design and development projects.	01	L_2	CO4
4	What is meant by "design parallelism" in Concurrent Engineering?	01	L_2	CO3
5	What impact does Concurrent Engineering have on reducing design iterations and rework?	01	L_2	CO3
6	How does Industrial Design differ from Engineering Design?	01	L_2	CO3
7	Why is Biomechanics important in Ergonomics for product design?	01	L_2	CO3
8	How do designers use eye-tracking technology in Ergonomic studies?	01	L_2	CO4
9	How does the Fitts' Law apply to Ergonomic design in interactive products?	01	L_2	CO4
10	What is the importance of Grip Design in handheld product development?	01	L_2	CO4

PART – B

Sl. No.	Questions	M	BT	CO
1.	How does cross-functional team collaboration improve the efficiency of product development projects? Provide real world industry based examples.	10	L_4	CO3
2.	Explain how simultaneous product and process development helps improve efficiency in Concurrent Engineering. Discuss with two case examples.	10	L_3	CO3
3.	Explain how 3D Printing and Additive Manufacturing are transforming Industrial Design. Discuss advantages and limitations, Discuss using examples.	10	L_3	CO4
4.	Discuss how sports equipment design (e.g., running shoes, bicycles) integrates Industrial Design and Ergonomics for better performance. Specific product features covering performance, ergonomics and industrial design must be illustrated with appropriate sketches.	10	L_4	CO4
5.	Explain the role of sustainability in Industrial Design. How can sustainable design principles be integrated into product development? Give two case examples with supporting illustrative sketches.	10	L_4	CO4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Particulars	CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**19th February 2025**	**Maximum Marks**	**05 + 25**
Course Code	**21IM78**	**Duration**	**60 Mins**
Sem	**VII Semester**	**Improvement CIE**	

INNOVATION AND ENTREPRENEURSHIP

Note:
1. Answer all the Questions.
2. Use suitable sketches and tabular column wherever necessary.
3. Observe the marks split-up and answer accordingly.

PART – A

Sl. No.	Questions	M	BT	CO
1	Comment on the importance of Maximization with regard to pricing strategies.	01	L1	CO1
2	Provide two examples business that you can start now by using your existing skills and generate revenue with bootstrapping.	01	L4	CO2
3	Identify any two secondary revenue streams.	02	L1	CO2
4	Comment on how startups have used "Content Marketing" effectively to bring awareness to their offerings with an example.	01	L3	CO3

PART – B

Sl. No.	Questions	M	BT	CO
1.	Write short notes on i. Segmentation and Targeting ii. Importance of Early adopters	5	2	2
2.	Suggest a solution demo for any two of the following companies: i. A company building houses on wheels ii. A company manufacturing grocery bags that dissolve in water iii. A company that has made an app for visually challenged people iv. A company that rents furniture	2.5 *2 = 5	3	3
3.	Explain how Minimum Viable Product helps in Potential Risk reduction.	5	3	4
4.	Taking an example of Cirque De Sole or any other example of your choice, Illustrate the 4-action framework by plotting strategy canvas to take advantage of uncontested market.	5	3	4
5.	What are the key benefits of using social media marketing for brand engagement?	5	2	4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max Marks	Quiz	01	03	01	--	--	01	01	01	02	--	--
		Test	--	05	12	08	--	--	05	05	05	10	--

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**18th February 2025**	**Maximum Marks**	**10 + 50**
Course Code	**21IM73G1**	**Duration**	**120 Mins**
Sem	**VII Semester**	**Improvement CIE**	

DATA MINING AND PREDICTIVE ANALYTICS (Elective)

Note:
1. Answer all the Questions.
2. Avoid verbose answers and answer specifically and to the point.

PART – A

Sl. No.	Questions	M	BT	CO
1	Why are distance metrics important in clustering algorithms?	01	L_2	CO1
2	What is a dendrogram, and how is it used in hierarchical clustering?	01	L_2	CO1
3	What is the significance of "random" in Random Forest?	01	L_2	CO1
4	Can Random Forest be used for both classification and regression tasks?	01	L_2	CO1
5	How does Gradient Boosting differ from Random Forest?	01	L_2	CO2
6	What is the role of the learning rate in Gradient Boosting?	01	L_3	CO2
7	How does anomaly detection differ from outlier detection?	01	L_3	CO2
8	Describe one advantage of using unsupervised learning for anomaly detection.	01	L_1	CO2
9	Name one advantage of using ensemble methods over a single predictive model.	01	L_1	CO2
10	What does the term "bagging" refer to in ensemble techniques	01	L_1	CO2

PART – B

Sl. No.	Questions	M	BT	CO
1.	Consider five data points in 1D: A=1 A = 1, B=3 B = 3, C=4 C = 4, D=8 D = 8, and E=10 E = 10. i. Compute the pairwise Euclidean distances between these points. ii. Using single linkage, perform agglomerative clustering until you have two clusters. List the clusters at each step. *Hint:* In single linkage, the distance between two clusters is the minimum distance between any pair of points in the two clusters.	10	L_2	CO1
2.	You are building a decision tree for a binary classification problem where you have the following data at a node: Class: 0 — Count: 30; Class: 1 — Count: 70 i. Calculate the Gini impurity for the current node. ii. Suppose the dataset is split into two groups, one with 10 class 0 and 30 class 1, and the other with 20 class 0 and 40 class 1. Calculate the Gini impurity for each of the two splits and compute the weighted average Gini impurity for the entire split.	10	L_2	CO2
3.	Suppose you have a binary classification problem with the following dataset: x1: 2, 1, 3, 4 — x2: 3, 4, 2, 3 — y: 0, 1, 0, 1	05	L_3	CO2

	You decide to use Gradient Boosting Classifier with decision stumps as base learners. For simplicity, let's use a learning rate of η=0.1 and fit one shallow tree (stump) at each boosting iteration.		
4.	Describe the concept of Text Mining and how it is applied in extracting meaningful information from large volumes of unstructured text data. Provide an example of a text mining application.	15	L4
5.	You are given a dataset containing the transaction details of a credit card company. The dataset includes the following features: transaction amount, transaction time, transaction location, and the customer's previous transaction history. Define what anomaly detection is and explain its importance in the context of credit card fraud detection. Give one more example based on the above description.	10	L3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4
	Max Marks	Quiz	05	05	--	--	--	02	06	02	—
		Test	10	40	--	--	--	--	35	05	10

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

Date	18th February 2025	Maximum Marks	10 + 50
Course Code	21IM73GC	Duration	120 Mins
Sem	VII	Improvement CIE	

VIRTUAL COMMERCE (Department Elective)

Note: Answer all the Questions.

PART – A

Sl. No.	Questions	M	BT	CO
1	What are the key limitations of the current internet infrastructure in meeting the increasing demands of modern society?	02	L2	CO1
2	How can decline costs of IoT devices be applied to maintain reliability and minimize error rates in a large-scale deployment?	02	L2	CO2
3	What is the role of internet access drones in improving internet connectivity, particularly in remote areas?	02	L1	CO3
4	What is the impact of fiber optics in the First Mile on the overall performance of internet services, considering bandwidth and scalability.	02	L2	CO2
5	Compare the strengths and weaknesses of mobile internet access vs. telephone-based internet in the Last Mile, focusing on reliability.	02	L2	CO3

PART – B

Sl. No.	Questions	M	BT	CO
1.	Analyze the limitations of the current internet infrastructure in handling the increasing demands of modern society. Evaluate how the introduction of Internet2® could address these limitations. Furthermore, assess the impact of government regulation and surveillance on the internet, considering both societal benefits and privacy concerns. How do these regulations influence the development of future internet technologies, and what balance should be struck between security and privacy?	10	L4	CO4
2.	Analyze the challenges and advantages of deploying fiber optic technology in the First Mile for handling the bandwidth explosion. How does it impact the scalability and reliability of internet infrastructure? In the context of the Last Mile, compare and evaluate mobile internet access versus telephone-based internet access in terms of speed, accessibility, and reliability. How do these two access technologies influence the overall performance and equity of global internet connectivity?	10	L4	CO4
3.	imagine you are tasked with improving internet latency for a global online gaming platform. Apply your knowledge of future internet latency solutions (such as edge computing, 5G, or satellite networks) to propose a set of strategies that could reduce latency and improve the user experience. How would you prioritize and implement these strategies in a high-traffic gaming environment?	10	L3	CO3
4.	Given the increasing deployment of IoT devices in smart homes and industrial settings, apply your knowledge of guaranteed service levels and lower error rates to design an IoT-based solution that ensures optimal performance and reliability. How would you implement a system that balances low error rates with maintaining the cost-effectiveness of the solution, considering the declining costs of IoT devices?	10	L3	CO1
5.	How has Social Commerce evolved over time? What are its advantages and disadvantages?	10	L2	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max Marks	Quiz	05	05	--	--	--	04	06	--	--	--	--
		Test	10	10	10	20	--	10	20	10	10	--	--

Academic year 2024-2025 (ODD Sem

DEPARTMENT OF

INDUSTRIAL ENGINEERING & MANAGEMENT

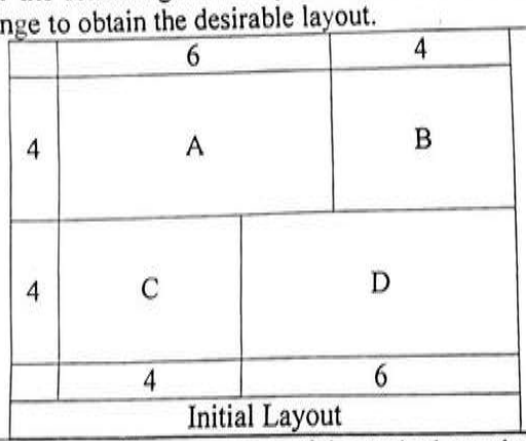
Date	18th February 2025	**Maximum Marks**	10+50
Course Code	21IM74HA	**Duration**	120 Min
Sem	VII	**Improvement CIE**	
FACILITIES PLANNING AND DESIGN			

Note:

1. Answer all the Questions.

	Part – A			
Q.No	**Questions**	**M**	**BT**	**CO**
1.1	List any two basic information required for ALDEP algorithm.	(02)	L1	CO1
1.2	The mission of a warehouse is to effectively _____ products in any configuration to the next step in the supply chain without _____ or altering the product's basic form.	(02)	L2	CO1
1.3	List any two Receiving Principles.	(02)	L1	CO1
1.4	State any two limitations of CRAFT?	(02)	L1	CO1
1.5	Activity relationship provides the basis for __ in the facilities planning ___.	(02)	L1	CO1

	Part –B			
Sl. No.	**Questions**	**M**	**BT**	**CO**
1	Consider the following initial layout with unit cost matrix. Use the CRAFT pair wise interchange to obtain the desirable layout.	10	L3	CO3
2	Illustrate and provide a summary of the typical warehouse functions and flows.	10	L3	CO2
3	State and explain the steps involved in Automated Layout Design Program (ALDEP).	10	L2	CO3
4	Explain the steps in designing of layout corresponding to typical U Flow for a manufacturing facility.	10	L3	CO4
5	With suitable assumptions propose a layout plan for setting up facilities for an information technology organization.	10	L4	CO4



Initial Layout

	A	B	C	D
A	-	1	1	3
B	2	-	2	1
C	0	2	-	2
D	1	4	3	-

Flow Matrix

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

	Particulars		**CO1**	**CO2**	**CO3**	**CO4**	**L1**	**L2**	**L3**	**L4**	**L5**	**L6**
Marks Distribution	**Quiz**	**Max Marks**	10	-	-	-	08	02	-	-	-	-
	Test		10	20	20	-	10	30	10	-	-	-

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Academic year 2024-2025 (Odd Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**19th February 2025**	**Maximum Marks**	**10+50**
Course Code	**21IM74HD**	**Duration**	**120**
Sem	**VII**	**Improvement CIE**	

DATABASE MANAGEMENT SYSTEM

Note: Answer all the Questions.

Sl No	Questions	M	BT	CO
	Part A			
1.	Identify the basic data types available for attributes in SQL.	(02)	L2	CO1
2.	Mention the guidelines for physical database design.	(02)	L2	CO3
3.	Distinguish between the base relation and virtual relation in SQL environment.	(02)	L3	CO4
4.	"Insertion, deletion and modification anomalies are considered bad in database design". Justify.	(02)	L5	CO1
5.	Mention the expended form of the acronym BCNF.	(02)	L1	CO3
	Part B			
1	Why is Normalization necessary while designing the relational data base? Analyze various levels in Normalization process.	(10)	L4	CO1
2	Write SQL commands to perform the following: i) To create a table student with fields: USN, SName, Address, Course, Section, TotalMarks ii) To insert values in the above table interactively iii) To create view with fields USN & Name. iv) To change the total marks of 50 if the marks lie in between 47 & 49. v) To delete tuples from the relation if the total marks is less than 50.	(10)	L3	CO1
3	Discuss insertion, deletion, and modification anomalies. Why are they considered bad? Illustrate with examples.	(10)	L4	CO3
4	Consider the following relations for an order-processing application database at ABC, Inc. ORDER (O#, Odate, Cust#, Total_amount) ORDER_ITEM(O#, I#, Qty_ordered, Total_price, Discount%) Assume that each item has a different discount. The Total price refers to one item. Odate is the date on which the order was placed, and the Total_amount is the amount of the order. If we apply a natural join on the relations ORDER_ITEM and ORDER in this database, what does the resulting relation schema look like? What will be its key? Show the FDs in this resulting relation. Is it in 2NF? Is it in 3NF? Why or why not?	(10)	L3	CO4
5 a	Give an overview of different types of JOINS available in SQL.	(06)	L3	CO3
b	Distinguish between JOIN and CARTESIAN PRODUCT.	(04)	L4	CO3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

	Particulars	CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
Marks Distribution	Quiz Max Marks	04	-	4	2	-	02	04	02	-	02	-
	Test	20	-	20	10	-	-	-	26	24	-	-

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DEPARTMENT OF

BIOTECHNOLOGY

Academic year 2024-2025 (Odd Sem)

Common to all branches

Date	February, 2025	Maximum Marks	10 + 50
Course Code	21BT75IB	Duration	120 min
Sem	VII Semester	Closed Book Offline Quiz and Test – 2	

Healthcare Analytics (Institutional Elective)

Answer all questions in Part A & Part B. Answer Part A in first two pages in sequence only

Part A

Sl No	Questions	M	BT-L	CO
1	Write any 2 applications of systems biology.	2	2	1
2	Mass Spectrometry measures _____charge ratio (m/z) of molecules	1	2	1
3	Mention any 2 steps involved in implementation of systems biology.	2	2	2
4	List any 2 applications of secondary structure prediction.	2	3	2
5	What are the 2 gene prediction methods?	2	2	3
6	Name any tool used to detect the functional sites in a gene	1	4	3

Part B

Sl. No	Questions (Test)	M	BT-L	CO
1.	Elucidate on the gene prediction methods	10	2	3
2	How does the RNA secondary be predicted?	10	3	3
3	Explain the basic structure of a protein.	10	4	4
4	Illustrate on the protein structure predictions.	10	4	4
5	Give a comprehensive note on the protein identity based on the composition.	10	2	3

BTL-Blooms taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks			60	40		40	20	40		

Academic year 2024-2025 (Odd Sem)

DEPARTMENT OF

CHEMICAL ENGINEERING

Date	Feb 2025	Maximum Marks	60
Course Code	21CH751C	Duration	120 Min
Sem	VII Semester	CIE 3	

SUSTAINABILITY AND LIFE CYCLE ANALYSIS (Institutional Elective)

Note: 1. Answer all the Questions.

	Part A	M	BT	CO
a	Define the terms odour character and odour persistence.	2	1	2
b	How do you control the odour arising from hydrogen sulphide?	2	1	2
c	Name two fermentative bacteria	2	1	3
d	Write a relation between intensity and concentration,	2	1	3
e	Name two bacteria in acetogenesis.	2	1	2
	Part B			
	Explain the working of atomisers to control the odour	10	2	3
	State and Explain the stages of biomethanation	10	2	4
	With a neat sketch explain the working of fixed film digestion system	10	2	3
	What are the stages of bioethanol production? Explain in brief	10	2	3
	With a neat diagram explain the working of Down draft Gasifier	10	2	4

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Academic year 2024-2025 (Odd Sem)
(IMPROVEMENT TEST FOR VII SEM)

DEPARTMENT OF CHEMISTRY

Date	19.2.2025	Sem - VII	Improvement test
Course Code	21CM75ID	Maximum Test Marks	Quiz-10 + Test-50 (60M)
Course Name	ACSM	Duration	120 mins

ADVANCES IN CORROSION SCIENCE AND MANAGEMENT

Note: 1) Answer all the questions
2) Part A should be answered in the first two pages of the answer script.

	Part-A	M	BTL	CO
1	Name the type of corrosion that occurs commonly in oil and gas industries.	1	1	2
2	Arrange the following metals and alloys according to their position in the galvanic series: Gold, Aluminum, Copper, and Magnesium alloy, Bronze Gold, Aluminum, Copper and Magnesium alloy, Bronze.	1	2	2
3	Suggest any one method to reduce seasonal cracking.	1	1	1
4	What is the consequence of hydrogen embrittlement?	1	2	3
5	Pitting is localized and has intense corrosion. Justify.	1	3	2
6	State any one favourable condition required for rusting to take place.	1	2	1
7	List any one disadvantage of a corrosion map.	1	1	2
8	What is the significance of the Tafel equation in corrosion studies?	1	3	2
9	How does electrical isolation help prevent differential metallic corrosion?	1	2	2
10	List any one preventive measure to avoid microbial corrosion.	1	1	3

	Part-B	M	BTL	CO
11	What are the consequences and effects of corrosion, explain with relevant examples.	7	2	2
12	What is the significance of creating a corrosion map of India, and how can it help in addressing corrosion issues more effectively?	7	3	3
13	What are the necessary steps to be taken for the corrosion test program? Explain	7	2	1
14	Two metals A and B undergo oxidation corrosion in the absence of moisture. A ...des faster and corrosion of B takes place negligibly. Predict their Pilling-	7	3	4

Academic year 2024-2025 (Odd Sem)
(IMPROVEMENT TEST FOR VII SEM)

	Bedworth ratio and give one example for each. Also, state the Pilling-Bedworth rule.	
15	Differentiate between the Galvanic series and the electrochemical series, highlighting the key distinctions in their representations and significance	
16	Explain the following types of corrosion i) Crevice corrosion ii) Erosion corrosion	
17	Enumerate four benefits and constraints associated with corrosion testing techniques.	8

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	BTL	
	Max Marks	16	13	24	07	L1	L2
						4	33

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Academic year 2024-2025 (Odd Sem)
(IMPROVEMENT TEST FOR VII SEM)

	Bedworth ratio and give one example for each. Also, state the Pilling-Bedworth rule.	
15	Differentiate between the Galvanic series and the electrochemical series, highlighting the key distinctions in their representations and significance	7
16	Explain the following types of corrosion i) Crevice corrosion ii) Erosion corrosion	7
17	Enumerate four benefits and constraints associated with corrosion testing techniques.	8

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	BTL		
	Max Marks	16	13	24	07	L1	L2	L
						4	33	3

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Department of Computer Science and Engineering

Academic year 2024-2025 (Odd Sem)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Date	Feb 2025	Maximum Marks	50+10
Course Code	21CS75IE	Duration	120 Min
Sem	07	Improvement CIE	

PROMPT ENGINEERING
(INSTITUTIONAL ELECTIVES – II (Group G)))

Q. No.	Questions	M	BT	CO
	QUIZ			
1.	1.1 Role-playing prompts often begin with instructions to the model to act as a specific _____ or character.	01	1	1
	1.2 Which setting adjusts the probability distribution to control randomness in the generated text?	01	2	2
	1.3 Why is providing context in a prompt important, and how does it help the model?	01	2	2
	1.4 What is the significance of providing examples in prompts for complex tasks?	01	1	1
	1.5 What is one method to prevent prompt injection in AI models?	01	1	1
	1.6 How does prompt leaking impact AI security?	01	1	1
	1.7 Name one strategy to prevent AI jailbreaking.	01	1	2
	1.8 Why is human feedback important in reinforcement learning?	01	2	3
	1.9 What role does reward function play in reinforcement learning?	01	1	1
	1.10 Name one key algorithm used in reinforcement learning for planning.	01	1	1
	TEST			
2.	(a)Describe an example where PAL can enhance the performance of a language model.	05	2	3
	(b)Explain the concept of chain-of-thought (CoT) prompting and its significance in reasoning tasks.	05	1	2
3	Classify and explain the types of Large Language Model (LLM) prompts based on the type of information being prompted. Provide examples for each type.	10	1	2
4	What is multimodal prompting in AI models? Discuss its significance and	10	1	2

challenges in integrating different types of data (text, images, audio, etc.).

5 Define graph prompting and discuss its role in enhancing reasoning capabilities in language models.

6 How can adversarial attacks such as prompt injection and jailbreaking impact the security of AI systems? Suggest possible mitigation strategies.

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BT-Blooms Taxonomy, CO-Course Outcomes

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4
	Test & Quiz	Max Marks	06	28	16	00	10	32	18	00	11

Course Outcomes:	
CO1:	Demonstrate an understanding of prompt engineering principles including how prompt structure and phrasing impact the performance of AI models.
CO2:	Design and implement effective prompts- to create and apply prompts for various natural language processing (NLP) tasks, such as text generation, summarization, and translation AI models.
CO3:	Critically evaluate the effectiveness of prompts - assess the quality and performance of in terms of accuracy, coherence, and relevance, identifying areas for improvement.
CO4:	Apply prompt engineering techniques in real-world scenarios - use prompt engineering strategies to address practical problems in domains such as education, healthcare, and demonstrating the applicability of AI-driven solutions.
CO5:	Collaborate on projects involving prompt engineering - work effectively in teams to design, implement, and evaluate prompt-based solutions, showcasing their ability to contribute to complex AI-related projects.

Department of Civil Engineering

Date	19-02-2025	Maximum Marks	50+10
Course Code	21CV75IF	Duration	120 Min
Sem	VII	IMPROVEMENT CIE	
INTEGRATED HEALTH MONITORING OF STRUCTURES			

Note: Answer all questions.

QUIZ

Sl. No.	Questions	M	BT	CO
1	Sensors measure physical parameters such as _____, _____, _____ during testing. (any 2)	2	1	2
2	Two main types of Dynamic Analysis are _____ and _____	2	1	4
3	List any two characteristics of Static Response Measurement	2	1	3
4	Mention any two advantages of Dynamic Response methods.	2	1	4
5	List the different NDT methods (any 4)	2	1	4

TEST

Sl. No.	Questions	M	BT	CO
1	Explain the objectives of Static response systems and Discuss the measurement techniques used for static response systems..	10	2	3
2	List and discuss the different types of Dynamic Response Methods	10	2	3
3	Discuss different remote SHM techniques, systems and challenges	10	2	4
4	Describe remote data acquisition systems and discuss the components used	10	3	4
5	Explain different Simulation methods and loading methods of Static Field Testing.	10	3	3

Course outcomes:

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	-	2	32	26	10	30	20			

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Department of Electrical & Electronics Engineering

Academic Year 2024-25 (ODD SEM)

Date	19th Feb 2024	Maximum Marks	50
Course Code	21EE75IH	Duration	90 Mins
Sem	VII Semester	TEST-III	
E-Mobility			

Q.No	Part B – Test Questions	Marks	COs	BT
1a.	Draw and explain the functions of components involved in Low power charger	06	4	2
1b.	Mention voltage and current levels of AC and DC charging.	04	2	1
2.	With the neat sketch of circuit diagram and waveforms explain the operation of Boost Converter for Power Factor Correction.	~~06~~ 10	4	3
3a.	Write the various EV charging standards and discuss briefly about ***SAE J1772***.	05	2	2
3b.	Discuss briefly about wireless charging of electric vehicle.	05	2	2
4.	Discuss the following with respect considerations for sizing of drive system. (a) Sizing the propulsion motor (b) Matching the electric machine and the internal combustion engine (ICE).	10	4	2
5a.	Discuss briefly about typical CAN layout in electric vehicle.	05	3	3
5b.	Explain about CAN message frame.	05	3	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	-	14	10	26	4	26	16	-	-	-

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Department of Electrical & Electronics Engineering

Academic Year 2024-25 (ODD SEM)

Date	19th Feb 2024	Maximum Marks	
Course Code	21EE75IH	Duration	
Sem	VII Semester		QUIZ-iii

E-Mobility

Q.No.	Part A – Quiz Questions	Marks	CO
1.	What are the key requirements of an EV charging system?	02	2
2.	Why is power factor correction important in battery charging?	02	2
3.	What is the function of an AC-DC converter in an EV charger?	02	2
4.	What is the role of a communication protocol in EV charging?	02	3
5.	What role does an onboard charger play in an EV?	02	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4
	Test	Max Marks	-	06	04	-	06	03	01	-

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Department of Electronics & Telecommunication Engineering

Date	19th FEB. 2025	Maximum Marks/Quiz	50+10
Course Code/Title	21ET75IK	Duration	120 Min
Sem	VII Semester	CIE-III	

SPACE TECHNOLOGY AND APPLICATIONS

Q.No	PART A	M	BT	CO
1.	Define Multispectral Image.	1	L1	CO3
2.	What is Spatial and Spectral resolution?	1	L2	CO3
3.	Mention few applications of Remote sensing	1	L1	CO4
4.	Who was the first person to walk on the moon?	1	L1	CO4
5.	Define Space debris.	1	L1	CO4
6.	What is the difference between space shuttle and spacecraft?	1	L2	CO4
7.	Differentiate between the US Apollo Mission and the India Chandrayaan -1 mission?	2	L2	CO4
8.	What are the most common applications of satellite in day to day life? Mention any four.	2	L2	CO1
	PART B			
1.	Explain how image processing and analysis applied in remote sensing. Illustrate relevant techniques used.	10	L2	CO3
2.	Explain different types of remote sensing cameras by mentioning their various characteristics and differences.	10	L2	CO3
3.	Describe at least two remote sensing satellites launched by India and highlight their applications.	10	L2	CO4
4.a)	What are the Procedures adopted for Disaster and flood management system and explain with the help of a neat diagram.	05	L2	C3
b)	Write a short note on "Inter space communication system"	05	L2	C3
5.	Give explanatory notes on: i) Planetary Payloads ii) Zero gravity experiments.	10	L2	CO4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	01	___	32	17	06	44	---	--	--	--

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**19 FEB 2025**	**Maximum Marks**	**65**
Course Code	**21IM75IM**	**Duration**	**120 Min**
Sem	**VII**	**Improvement Test**	

PROJECT MANAGEMENT

Note:
1. Answer all the Questions.

PART – A

Sl. No.	Questions	M	BT	CO
1	Which costs are included in project life cycle costing?	02	L2	CO1
2	What are the four main stages of Product Lifecycle Management (PLM)?	02	L2	CO1
3	What is resource levelling in project scheduling?	02	L2	CO2
4	What are the two major network-based scheduling techniques?	02	L2	CO1
5	What is project implementation scheduling?	02	L2	CO2

PART – B

Sl. No.	Questions	M	BT	CO
1.	A manufacturing company is launching a new product and needs to plan its project schedule efficiently. As a project manager, explain the steps involved in project implementation scheduling and discuss how effective time management and resource allocation methods can optimize the project timeline.	10	L3	CO3
2.	A construction company is implementing a new infrastructure project. How can different scheduling techniques and PLM (Product Lifecycle Management) concepts be used to optimize project planning? Additionally, explain the significance of life cycle costing in project management.	10	L3	CO3
3.	A project consists of the following activities with their estimated durations (in days) and dependencies:	15	L4	CO4

Activity	Predecessor(s)	Duration (Days)
A	-	5
B	A	4
C	A	6
D	B	3
E	B, C	7
F	D, E	5

Tasks:
1. Construct the Project Network Diagram.
2. Identify the Earliest Start Time (ES), Earliest Finish Time (EF), Latest Start Time (LS), and Latest Finish Time (LF) for each activity.
3. Determine the critical path and the total project duration.

4. A project consists of the following activities with their estimated time durations in weeks:

Activity	Predecessor	Optimistic (a)	Most Likely (m)	Pessimistic (b)
A	-	2	4	6
B	A	3	6	9
C	A	4	5	6
D	B	2	3	8
E	C	3	4	5
F	D, E	4	6	8

Tasks:

1. Construct the Project Network Diagram.
2. Compute the Expected Time (TE) for each activity
3. Identify the Critical Path and the Project Completion Time.

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3
	Max Marks	Quiz	6	4					10	
		Test			20	30				20

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DEPARTMENT OF MECHANICAL ENGINEERING

Date	19/02/2025	Max. Marks	10 (Q) + 50 (T)
Course Code	21ME75IO	Duration	120 Min.
Semester	VII	CIE-III (Improvement Quiz and Test)	

NUCLEAR ENGINEERING

Answer all the Questions.

#	Questions	M	BT	CO
	Part – A (Quiz)			
1	A nuclear unit becoming critical means	01	L1	01
2	Moderator in nuclear plants is used to	01	L1	01
3	The most commonly used moderator in nuclear plants is	01	L1	01
4	Breeder reactor has a conversion ratio of	01	L2	01
5	One gram of uranium will produce energy equivalent to approximately .....tonne of high-grade coal.	01	L2	01
6	The commonly used material for shielding is	01	L1	01
7	The main interest of shielding in nuclear reactor is protection against	01	L1	01
8	Reflector in nuclear plants is used to	01	L1	01
9	The energy required to be applied to a radioactive nucleus for the emission of a neutron is....	01	L2	01
10	Enriched uranium is one in which	01	L2	01
	Part – B			
1	Explain with a sketch the elements of a nuclear reactor.	10	L3	02
2	Explain the factors that control the site selection of nuclear power plant.	10	L4	02
3	Give the classification of nuclear reactors using various parameters.	10	L5	03
4	A 230 g piece of boron (mol wt. 10) absorbs thermal neutrons at the rate of 9.57×10^{13} per (cm^3 s). Boron density is 2.3 g/cm^3. Find (a) the thermal neutron flux, and (b) the average distance that a neutron travels before it is absorbed. For thermal neutrons, σ_a = 755 barns and σ_s = 4.0 barns.	10	L5	03
5	A reactor is fuelled with 100 tonnes of natural uranium (atomic mass 238.05) in which the average thermal neutron (2200 m/s) flux is 10^{13} neutrons/cm^2s, The 2200 m/s cross section of U-235 (atomic mass 235.04) are: σ_f = 579 barns and σ_c = 101 barns. The energy release per fission is 200 MeV and 0.715% of natural uranium is U-235. Calculate (a) the rating of the reactor in MW/tonne, (b) the rate of consumption of U-235 per day.	10	L6	04

BT-BloomsTaxonomy,CO-CourseOutcomes,M-Marks

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	MaxMarks	10	20	20	10	06	04	10	10	20	10

*****All The Best*****

DEPARTMENT OF

Humanities and Social Sciences

Date	19th February 2025		**Maximum Marks**	10 + 50
Course Code	21HS75IQ		**Duration**	30 + 90 Min
Sem	VII		**CIE – III- Improvement Test**	

Cognitive Psychology

Note: Answer all the Questions.

	Part – A- Quiz	**10 Marks**		
Sl. No.	**Questions**	**M**	**BT**	**CO**
1.1	What properties characterize language?	2	L1	CO1
1.2	What is the difference between phonemes and morphemes?	2	L1	CO1
1.3	Which processes can be impaired in dyslexia?	2	L1	CO2
1.4	What are the fundamental structures and processes within the brain?	2	L2	CO2
1.5	Why is the study of brain disorders useful for cognitive psychologists?	2	L2	CO2
	Part – B	**50 Marks**		
Q. No	Questions	M	BT	CO1
2	Compare and contrast the kinds of understandings that can be gained by studying speech errors made by healthy people with those that can be gained by studying the language produced by people who have particular brain lesions.	10	L1	CO1
3	Explain in detail how cognitive psychologists should interpret evidence of linguistic universals when considering the linguistic-relativity hypothesis.	10	L2	CO2
4	Researchers already are aware that a deficit of a neurotransmitter, acetylcholine, in the hippocampus is linked to Alzheimer's disease. Given the difficulty of reaching the hippocampus without causing other kinds of brain damage, how might researchers try to treat Alzheimer's disease?	10	L3	CO2
5	What is an area of cognition that could be studied effectively by viewing the structure or function of the human brain? Describe how a researcher might use one of the techniques to study that area of cognition.	10	L1	CO1
6	Explain some of the reasons that researchers are interested in finding out the localization of function in the human brain.	10	L2	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	**Particulars**		**CO1**	**CO2**	**CO3**	**CO4**	**L1**	**L2**	**L3**	**L4**	**L5**	**L6**
	Test	**Max Marks**	10	20	20	--	20	20	10	--	--	--

Academic year 2024-2025 (ODD SEM)

DEPARTMENT OF

INDUSTRIAL ENGINEERING& MANAGEMENT

Date	19/02/2025	Maximum Marks	10+50
Course Code	21IHS75IR	Duration	120 Mins
Sem	VII	IMPROVEMENT TEST	
PRINCIPLES & PRACTICES OF CYBER LAW			

Note: Answer all the Questions.

Q.No.	PART- A	M	BT	CO
1	Name the two different types of data with their meaning	2	3	CO2
2	Define Meta-Data.	2	2	CO3
3	What are Forensic Tools ?	2	2	CO2
4	What is an FIR in the context of Cyber Crime Investigation?	2	2	CO2
5	What is a Charge Sheet in Cyber Crime Investigation?	2	2	CO2
	PART -B			
1	Define Cyber Crime Cells (CCCs) and discuss their role in the investigation of cybercrimes.	10	02	CO3
2	Explain the structure and key functions of Cyber Crime Cells in India.	10	02	CO3
3	Explain the process of cyber crime investigation in detail, including the role of law enforcement agencies, evidence collection, and legal procedures.	10	03	CO3
4	Discuss the need to protect data in cyberspace and the types of data that require protection.	10	03	CO2
5a	Discuss the General Data Protection Regulations (GDPR) 2016.	05	03	CO2
5b	How can organizations ensure compliance with data protection regulations?	05	03	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	-	28	32	-	-	18	42	-	-	-

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DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

24th January 2023	Maximum Marks	100
AIMS 1	Duration	120 Mins
VII Semester	OE – II	

CONSTITUTION OF INDIA AND PROFESSIONAL ETHICS

Answer All the Questions

PART – A

Questions	M	BT	CO
Explain Doctrine of Civil Code	02	2	1
Write any two powers of National Human Rights Commission (NHRC)	02	2	1
Explain the concept of product liability	02	2	1
Distinguish between unfair Trade practice & Deficiency in Services	02	2	1
Write any two consumer Rights available under the Act	02	2	1

PART – B

	Questions	M	BT	CO
2 a	Examine the Constitutional position of Governor of a State	05	2	1
2 b	Discuss Online shopping and product liability covering-Liability of a product retailer of delivery service and Liability related to online food products.	05	2	1
3 a	Explain the Impeachment proceedings of President of India.	05	2	2
3 b	Discuss the ground and procedure of removing the Judges of Supreme Court	05	3	2
3	List 11 Fundamental Duties enshrined in the Constitution of India.	10	2	3
4	The slogan line of "Jago Grahak Jago" is applicable today. Discuss the Historical perspective, Significance and Key features of Consumer Protection Act (both old and new).	10	3	
5	Directive principle is fundamental in making Welfare state". Discuss	10	4	3

BT-Bloom's Taxonomy, CO-Course Outcomes, M-Marks

Mark Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max Marks	Quiz	--	10	--	--	--	--	10				
		Test	20	20	10	-	25	15	10				